CHE 311H - Fall 2023

HW #7

Due: November 28, 2023, during class. Submit as a hardcopy.

For full credit on all problems, make sure to indicate which figure from Geankoplis you used to obtain each Y value

Problem 1: A thermocouple junction, which may be approximated as a sphere, is to be used for temperature measurement in a gas stream. The heat transfer coefficient between the junction surface and gas is given by h = 400 $W/m^2.K$, and the junction thermophysical properties are k = 20 W/m.K; Cp = 400 J/kg.K and ρ = 8500 kg/m^3.

a) Determine the junction diameter needed for the thermocouple to have a time constant of 1 s. The time constant is defined as $\tau = (\rho.V.Cp)/(A.h)$, where V and A represent volume and area respectively.

b) If the junction is at 25 °C and is placed in a gas stream that is 200 °C, how long will it take for the junction to reach 199 °C?

Problem 2: In laying water mains, utilities must be concerned with the possibility of freezing during cold periods. Although the problem of determining the temperature in soil as a function of time is complicated by changing surface conditions, reasonable estimates can be based on the assumption of a constant surface temperature over a prolonged period of cold weather. What minimum burial depth x would you recommend to avoid freezing under conditions for which soil, initially at uniform temperature of 20 °C, is subjected to a constant surface temperature of -15 °C for 60 days? Use the given properties of soil at 300 K: ρ = 2050 kg/m^3; k = 0.52 W/m.K; Cp = 1840 J/kg; α = 0.138x10-6 m^2 /s

Problem 3: In a manufacturing process, stainless steel cylinders initially at 600 K are quenched by submersion in an oil bath maintained at 300 K with h = 500 W/m^2 . Each cylinder is of length 2L = 60 mm and diameter D = 80 mm. Consider a time 3 minutes into the cooling process and determine the temperature at the center of the cylinder. Note that there is heat transfer from all surfaces and that the temperature within the cylinder is a function of both position and time. Properties of stainless steel at T = (600+300)/2 = 450 K: ρ = 7900 kg/m^3; k = 17.4 W/m.K; Cp = 526 J/kg; α = 4.19x10-6 m^2/s

Problem 4: Let us consider a spherical cantaloupe 10 cm in diameter that is taken from a 24 °C environment and placed in a refrigerator at 4 °C. Assume the thermal conductivity of the cantaloupe is k = 5 W/cm.K; α = 1.25x10-2 cm^2/s; and the heat transfer coefficient for air is h = 1 W/cm^2 .K.

a) Determine the temperature in the cantaloupe 1 cm radially out from the center after 50 minutes.
b) How long does the cantaloupe have to be in the refrigerator to bring its temperature 3 cm below the surface to 6 °C?

Problem 5: BSL 14A.9 You will need to look up thermal properties of air to solve this problem – make sure to report the source(s) from which you obtained the thermal properties.